

Equitable Quotients Across Five Towers

A Lean-Verified Spine for Quantum-Walk Search and Transfer,
with a Graphon Quasi-Infinite Limit

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Abstract

We organize a body of continuous-time quantum-walk results — perfect state transfer, uniform mixing, spatial search — around a single structural spine: an equitable partition of a Hermitian weighted graph, with its small finite quotient matrix as the universal “small object”. The spine ascends a five-tower hierarchy: sets, simple graphs, Hermitian weighted graphs (which includes chiral and magnetic graphs), operator algebras and quantum graphs, graphons, and finally a categorical framing in which the Quotient functor preserves filtered colimits.

The headline novelty is at Tower 4: we lift the Bachman–Fredette–Fuller–Landry–Opperman–Tamon–Tollefson theorem on perfect state transfer on quotient graphs to the graphon limit. Together with a categorical filtered-colimit-preservation theorem this gives a clean “quasi-infinite limit theorem”: a sequence of finite quantum-walk problems with bounded equitable quotients has its search/transfer behavior controlled by a single graphon quotient. The Xie–Tamon infinite-tail result is recovered as a one-line corollary.

We also recover the four-color-completion compiler of our earlier note as a single corner of a unifying GraphBundle construction, and we observe that chiral signing optimizations factor through the quotient: searching for a mixing-time speedup reduces to a finite signed problem on the template. The Lean 4 development covers the spine and the bundle engine; graphon, chiral, and categorical layers are scaffolded.

1. Introduction

The Tamon program has shown, over roughly two decades, that the right way to think about continuous-time quantum walks is through equitable partitions: the existence of perfect state transfer between two vertices of a graph is equivalent to the existence of perfect state transfer on the small quotient graph induced by any compatible equitable partition [bachman2011quotient]. Graph products are perfect-state-transfer constructions in disguise [ge2010products]. The sharpest known characterization of spatial-search optimality is also spectral, and is cleaner on quotients [chan2022shadows]. Finally, optimality survives unbounded one-dimensional probes [xie2023tail].

The previous version of this paper [amorc2025] reified a tiny slice of this program into a Lean 4 artifact and a Python compiler prototype, organized around the four-color completion of a planar graph. That version emphasized the role of a chosen color quotient: once a coloring is fixed there is a greatest graph compatible with it, and that greatest graph carries a Laplacian-integral search criterion via Li–Luo–Feng–Li [li2025deterministic].

This rewrite takes the same compiler stance but recognizes that color completion is one corner of a much larger construction. We organize results around a five-tower hierarchy, summarized by the diagram

$$\text{Set} \rightarrow \text{SimpleGraph} \rightarrow \text{WeightedGraph} \rightarrow \text{QuantumGraph} \rightarrow \text{Graphon} \rightarrow \text{Category}.$$

At every tower we maintain three load-bearing claims, all stated and (in the finite case) mechanized in Lean:

1. The quotient spectrum is contained in the original spectrum: $\text{spec}(A/\pi) \subseteq \text{spec}(A)$, with explicit eigenvector lifts.
2. Continuous-time quantum walk on the original graph restricts to the cell-constant subspace, with the same evolution as the quotient walk; PST, uniform mixing, and spatial search on cell-uniform initial states therefore lift from quotient to host.
3. At Tower 5 the Quotient functor preserves filtered colimits, which lets us push finite results through to graphon limits without re-proof.

Three results are genuinely new. First, the spine theorems work at Tower 4: we give graphon equitable partitions, prove the spectral-lift theorem for the Hilbert–Schmidt integral operator, and lift PST and spatial-search optimality to the graphon level. The closest ancestor is the Bick–Sclosa treatment of dynamical systems on graph limits [bick2024graphlimits], which gives an invariant-subspace view for real dynamics; we give the unitary version. Second, chiral signings of bundles can be optimized on the quotient: the recent uniform-mixing speedup of Levine et al. [levine2026chiral] fits naturally into this framework, and its optimization problem reduces to a small signed problem on the template. Third, the categorical filtered-colimit preservation theorem unifies the finite-stage results with the graphon limit, formalizing the slogan “quasi-infinite”.

The rest of the paper specializes. Section 2 fixes the Tower-2 spine; Section 3 introduces the constructive engine that recovers products, joins, color completions, and engineered template joins from a single definition. Section 4 climbs to Tower 3; Section 5 to Tower 4 and the headline limit theorem; Section 6 to the categorical Tower 5. Section 7 collects two specializations (chiral and relational). Section 8 returns to the engineering vision: a search/transfer compiler that emits a Lean-checkable certificate.

2. The Spine: Equitable Partitions and the Universal Lift (Tower 2)

A **weighted graph** on a finite vertex set V is a Hermitian matrix $A \in \mathbb{C}^{V \times V}$ with zero diagonal. The class includes simple graphs (via 0/1 adjacency), signed and chiral graphs (unitary phases off the diagonal), and weighted spin networks. The Lean carrier is `Graphplay/Weighted.lean`.

Definition 2.1 (equitable partition). A partition π of V into nonempty cells $C_1, \dots, C_r$ is **equitable** for A if for every pair of cells C_j, C_k the row sum

$$b_{jk} := \sum_{w \in C_k} A_{vw}$$

is independent of $v \in C_j$. The **quotient matrix** $A/\pi \in \mathbb{C}^{r \times r}$ has entries $(A/\pi)_{jk} = b_{jk}$. The **characteristic matrix** $P \in \mathbb{C}^{V \times r}$ has $P_{vj} = 1$ if $v \in C_j$ and 0 otherwise. This data lives in `Graphplay/Equitable.lean`.

The defining identity is $AP = P(A/\pi)$.

Theorem 2.1 (spectral lift). Let π be equitable for A . Then $\text{spec}(A/\pi) \subseteq \text{spec}(A)$, and for every eigenvector x of A/π with eigenvalue λ , the vector $Px \in \mathbb{C}^V$ is a (cell-constant) eigenvector of A with the same eigenvalue. (`Graphplay/Spectral.lean`)

Theorem 2.2 (PST lift). Let π be equitable for A and let C_a, C_b be cells with $|C_a| = |C_b|$. Identify cell-uniform states with the standard basis of $\mathbb{C}^r$. If perfect state transfer occurs from C_a to C_b in the quotient at time τ — that is, $|\langle e_b, e^{-i\tau A/\pi} e_a \rangle| = 1$ — then PST occurs between the cell-uniform states on A . This is the content of [bachman2011quotient], formalized as `Graphplay/PST.lean`.

Theorem 2.3 (search lift). For a marked vertex $w \in V$, let π_w be the refinement of π that splits the cell containing w into $\{w\}$ and its complement within the original cell. Then π_w is equitable for the

search Hamiltonian $H_\gamma = -\gamma A - e_w e_w^*$, and the unitary e^{-itH_γ} restricts to the (cell-uniform, $\{w\}$ -aware) subspace, which has dimension at most $r + 1$. The classical spatial-search optimality criterion [chakraborty2020spatial] sharpened by [chan2022shadows] therefore reduces to a small-matrix calculation on this subspace. The equitable-partition-for-search idiom is due to [ide2022equitable]. Lean: `Graphplay/Search.lean`.

Theorem 2.1 is classical Godsil. Theorems 2.2 and 2.3 are the load-bearing form we will lift through Towers 3, 4, 5. The Lean development covers all three at Tower 2.

Example 2.4. The complete multipartite graph $K_{n_1, \dots, n_r}$ has the color partition as an equitable partition, with quotient matrix $(A/\pi)_{jk} = n_k$ for $j \neq k$ and 0 otherwise. The spectrum is the spectrum of this small matrix together with internal zero modes, recovering the calculation of [amorc2025].

3. The Constructive Engine: Bundles and Their Corners

The universal construction at the spine is a bundle.

Definition 3.1 (graph bundle). A **graph bundle** `GraphBundle Q H B` is the data

$$Q : \text{WeightedGraph}(I), \quad H : \forall i \in I, \text{WeightedGraph}(V_i), \quad B : \forall (i, j) \text{ with } Q_{ij} \neq 0, V_i \times V_j \rightarrow \mathbb{C},$$

subject to a Hermitian compatibility $B_{ij}(a, b) = \overline{B_{ji}(b, a)}$. The **total graph** $\text{Tot}(Q, H, B)$ on $\coprod_i V_i$ is given by

$$\text{Tot}(x, y) = \begin{cases} H_{i(x,y)} & \text{if } x, y \text{ lie in the same fiber } V_i \\ Q_{ij} B_{ij}(x, y) & \text{if } x \in V_i, y \in V_j, i \neq j \end{cases}.$$

Lean: `Graphplay/Bundle.lean`.

Many earlier constructions are corners.

Proposition 3.2 (corners).

1. With $H_i = 0$ and $B_{ij} = 1$ on Q -edges, $\text{Tot}(Q, H, B)$ is the `TemplateJoin(Q, V)` of [amorc2025].
2. With $Q = K_2$, fibers H_1, H_2 arbitrary, and $B_{12} = 1$ on a chosen product subset, one recovers Cartesian, lexicographic, strong, and tensor products as cases. The PST-product theorems of [ge2010products] fall out as Tower-2 corollaries.
3. With $Q = K_C$ on a color palette and $H_i = 0$, one recovers the color completion `ColorCompletion c` of [amorc2025]; with $B_{ij} = 1$ this is complete multipartite.
4. With Q a join graph in the sense of [kirkland2023join] and arbitrary fibers, one recovers their join construction; their PST-on-joins characterization is then a Tower-2 search-lift statement.

Theorem 3.3 (equitable iff fibers regular and couplings biregular). The cell partition $\pi = \{V_i\}_{i \in I}$ is equitable for $\text{Tot}(Q, H, B)$ if and only if

1. each fiber graph H_i is regular (constant row sum r_i), and
2. each coupling B_{ij} is **biregular**: $\sum_{b \in V_j} B_{ij}(a, b)$ is independent of $a \in V_i$, and $\sum_{a \in V_i} B_{ij}(a, b)$ is independent of $b \in V_j$.

When these conditions hold, the quotient matrix is $(A/\pi)_{ij} = r_i \delta_{ij} + Q_{ij} |V_j| \beta_{ij}$ where β_{ij} is the constant row sum of B_{ij} .

Corollary 3.4 (spectral inheritance). For equal-fiber bundles ($|V_i| = m$ for all i , $H_i = 0$, $B_{ij} = 1$):

$$\text{spec}(\text{Tot}(Q, H, B)) = m \cdot \text{spec}(Q) \cup (\text{internal zero modes}).$$

In particular, the CNO spectral-ratio criterion [chakraborty2020spatial] of [chan2022shadows] is inherited by the bundle from the template.

The Lean artifact verifies Theorem 3.3 as `Bundle.equitable_iff_regular_biregular` and Corollary 3.4 as `Bundle.spec_equal_fiber`. The engineering reading is that the compiler’s certificate is exactly the pair of regularity and biregularity conditions in Theorem 3.3 — both decidable, both Lean-checkable.

4. Tower 3: Operator-Algebra and Quantum-Graph Upgrade

The next tower sits above weighted graphs. Let $\mathcal{A} \subseteq M_V(\mathbb{C})$ be the unital $*$ -subalgebra generated by A and the identity. An equitable partition π corresponds to an orthogonal projection $\Pi = P(P^*P)^{-1}P^*$ onto the cell-constant subspace; the equitability condition is exactly $\Pi A \Pi = A \Pi$, i.e. $\Pi \in \text{Comm}(A)$. Lean carrier: `Graphplay/QuantumGraph.lean`.

This is the right setting in which to discuss association schemes and quantum graphs. The Bose–Mesner algebra of a commutative association scheme is the maximally symmetric case; fractional revival on schemes [chan2019fractional] is a Tower-3 phenomenon that descends to Tower 2 via the characteristic projection of the scheme partition.

Definition 4.1 (quantum graph). A **quantum graph** on a finite-dimensional C^* -algebra $M \subseteq B(H)$ is a self-adjoint M -bimodule of $B(H)$. An **equitable coarsening** is a unital $*$ -subalgebra $M_0 \subseteq M$ such that the orthogonal projection onto the M_0 -invariant vectors commutes with the bimodule’s adjacency operator.

Theorem 4.2 (operator-algebra spine). Let S be a quantum graph on M with adjacency operator A_S , and let M_0 be an equitable coarsening with projection Π . Then $\Pi A_S \Pi$ is the adjacency operator of a smaller quantum graph S/M_0 on M_0 , and

$$\text{spec}(A_{S/M_0}) \subseteq \text{spec}(A_S).$$

Moreover e^{-itA_S} commutes with Π , so CTQW evolution on S restricts to the M_0 -invariant subspace and acts there as $e^{-itA_{S/M_0}}$.

Theorems 2.1–2.3 are the special case $M_0 = \text{diagonal subalgebra coarsened by } \pi$. Lean: `Graphplay/QuantumGraph.lean`. The fractional-revival characterization of [chan2019fractional] on Hamming schemes is a Tower-3 restatement of a Tower-2 search-lift on the Hamming scheme partition, and is visible in this language without re-proving.

5. Tower 4: Graphons and the Quasi-Infinite Limit Theorem

We now lift the spine to graphon kernels. This is the headline section.

A **Hermitian graphon** is a measurable function $W : [0, 1] \times [0, 1] \rightarrow \mathbb{C}$ with $W(x, y) = \overline{W(y, x)}$ and $W \in L^2$. The induced integral operator $T_W : L^2[0, 1] \rightarrow L^2[0, 1]$, $(T_W f)(x) = \int_0^1 W(x, y)f(y) \, dy$ is compact, self-adjoint, and Hilbert–Schmidt. The continuous-time quantum walk on W is the unitary group $U_{W(t)} := e^{-itT_W}$. Carrier: `Graphplay/Graphon/Basic.lean`, `Graphplay/Graphon/Hermitian.lean`.

Definition 5.1 (graphon equitable partition). A measurable partition $[0, 1] = \bigcup_{k=1}^r A_k$ is **equitable** for W if for all $j, k \in \{1, \dots, r\}$ the function

$$x \mapsto \int_{A_k} W(x, y) \, dy$$

is constant almost everywhere on A_j . Call that constant $\tilde{b}_{jk}$. The **graphon quotient matrix** $W/\pi \in \mathbb{C}^{r \times r}$ has entries $(W/\pi)_{jk} = \frac{\tilde{b}_{jk}}{\sqrt{|A_j| |A_k|}}$ when we normalize to the rescaled-indicator basis $u_k := |A_k|^{-1/2} \mathbb{1}_{A_k}$. Lean: `Graphplay/Graphon/Equitable.lean`. The closest finite ancestor is the Bick–Sclosa invariant-subspace decomposition [bick2024graphlimits]; we record the operator-theoretic version inspired by the LQR invariant-subspace treatment of [gao2020graphon].

Theorem 5.2 (graphon spectral lift). If π is equitable for W , then $\text{spec}(W/\pi) \subseteq \text{spec}(T_W)$, with eigenfunctions of T_W given by $\sum_k x_k u_k$ for eigenvectors x of W/π . The orthogonal complement of $\text{span}\{u_k\}$ is invariant under T_W and gives the (possibly infinite) fiber-internal spectrum. Lean: `Graphplay/Graphon/Spectral.lean`.

Theorem 5.3 (graphon PST and CTQW lift). For Hermitian W with equitable partition π , the unitary $U_{W(t)}$ preserves the finite-dimensional cell-constant subspace $V_\pi := \text{span}\{u_k\}$ and acts there as $e^{-itW/\pi}$. Consequently, if perfect state transfer between two cell-uniform states occurs in the quotient at time τ , it occurs in the graphon walk. This is a graphon-level generalization of [bachman2011quotient].

Theorem 5.4 (graphon spatial-search lift). Let $S \subseteq [0, 1]$ be a measurable marked set of positive measure. The search Hamiltonian $H_{W,\gamma,S} = -\gamma T_W - \text{Proj}_{L^2(S)}$ has the refined partition $\pi_S := \pi \boxtimes \{S, S^c\}$ as a (graphon) equitable partition, and the search dynamics restrict to a Hamiltonian of finite dimension at most $2r$ on V_{π_S} . The optimal-search criterion of [chan2022shadows] therefore reduces to a finite spectral computation on W/π_S .

Theorem 5.5 (quasi-infinite limit theorem). Let $(G_n, \pi_n)_{n \in \mathbb{N}}$ be a sequence of finite Hermitian weighted graphs equipped with equitable partitions, all of size at most r . Suppose:

1. The pixel-step graphons W_{G_n} converge to a Hermitian graphon W in the cut norm.
2. The cell-fraction vectors converge to a probability vector $(|A_1|, \dots, |A_r|) \in \Delta^{r-1}$ with all $|A_k| > 0$.
3. The normalized quotient matrices G_n/π_n converge to a Hermitian matrix $M \in \mathbb{C}^{r \times r}$ in Frobenius norm.

Then W admits a measurable equitable partition π realizing the limit cell fractions and with $W/\pi = M$. Moreover, every cell-uniform PST or spatial-search problem on (G_n, π_n) has a well-defined limit on (W, π) , and the limit’s value is the value of the finite Hermitian problem on M . As a special case, recover [xie2023tail]: the limit of K_n plus a path of length n is the disjoint union of a complete graphon and an infinite path, with a 2-cell equitable partition; the resulting 2×2 search problem is exactly the Farhi–Gutmann complete-graph search, and optimality is preserved. The transfer-side companion [bernard2022tails] is recovered by the analogous 2-cell partition.

Lean: `Graphplay/Graphon/Limit.lean` (currently scaffolded; the proof reduces to (a) cut-norm-implies-operator-norm convergence under bounded entries [szegedy2010spectral], (b) Frobenius convergence of the quotient matrices, and (c) continuity of finite spectral optimization problems).

Example 5.6 (cycle blowup). Fix a template weighted graph Q on r vertices. The sequence $\text{Tot}(Q, 0, J)$ with equal fibers of size n , viewed as pixel-step graphons, converges in cut norm to the block-constant graphon W_Q with cells of measure $1/r$ and value Q_{jk} in block (j, k) . The quotient at every stage is $n \cdot Q$ (up to a renormalization that absorbs into the time parameter), and Theorem 5.5 gives the limit as the finite search problem on Q . The compiler examples of [amorc2025] are exactly the finite stages of such a sequence.

6. Tower 5: Categorical Framing

Let EqWGr be the category whose objects are pairs (G, π) with G a finite Hermitian weighted graph and π an equitable partition, and whose morphisms $(G, \pi) \rightarrow (G', \pi')$ are pairs (φ, θ) where φ is a weight-preserving cell-block map and θ refines π to a partition compatible with φ . Let $\text{Mat}_{\text{fin}}(\mathbb{C})$ be the category of finite Hermitian matrices.

Define $\text{Quot} : \text{EqWGr} \rightarrow \text{Mat}_{\text{fin}}(\mathbb{C})$ by $\text{Quot}(G, \pi) = G/\pi$.

Theorem 6.1 (filtered-colimit preservation). The functor Quot preserves filtered colimits. Explicitly, given a filtered diagram (G_α, π_α) in EqWGr whose underlying weighted graphs form a sequence with a graphon colimit W (with respect to the cut metric) and whose quotient matrices have a finite-dimensional Frobenius limit M , the colimit pair (W, π) exists in the extended graphon-equitable category and $W/\pi = M$. Lean: `Graphplay/Categorical.lean`.

Theorem 6.1 unifies several earlier observations.

Corollary 6.2. The quasi-infinite limit theorem (Theorem 5.5) is the Tower-4 instance of Theorem 6.1. The Tower-1 inverse-limit and edge-union universal properties of [amorc2025] are Tower-1 and Tower-0 instances. The “failed reflector” observation — there is no left adjoint to “forget the coloring” — is recovered as the statement that the underlying-graph functor **does not** have a colimit-preserving left adjoint; the relative left adjoint along a chosen template Q does exist and is the bundle construction of Section 3.

7. Chiral and Relational Specializations

Chiral. A **unitary signing** of a bundle $\text{Tot}(Q, H, B)$ is a Hermitian phase decoration of off-diagonal blocks: $B_{ij} \rightarrow \sigma_{ij} B_{ij}$ with $|\sigma_{ij}| = 1$ and $\sigma_{ji} = \overline{\sigma_{ij}}$. Define the **phase-equitable** partition condition by requiring biregularity of the signed couplings. Lean: `Graphplay/Chiral.lean`.

Theorem 7.1 (chiral signing as optimization on the quotient). For a bundle with phase-equitable cell partition, the mixing-time functional

$$M(\sigma; t) := \max_{v, w} |U_{\sigma, t}(v, w)|^2$$

restricted to cell-uniform initial and final states equals the corresponding quantity computed for the signed quotient $\sigma_Q := (\sigma_{ij})$ acting on Q . In particular, optimizing σ over the (typically large) phase space of bundle signings reduces to optimizing σ_Q over the much smaller phase space $U(1)^{E(Q)}$, modulo gauge.

Corollary 7.2. The uniform-mixing speedup of [levine2026chiral] on K_n^σ has a bundle version: if a bundle has K_n^σ as its quotient and equal fibers, then the corresponding chiral mixing speedup holds for the bundle on the cell-uniform subspace. Open: characterize fiber-internal interference for the full uniform-mixing problem on the bundle.

Relational / CSP. A k -ary template is a relation $T \subseteq I^k$. Given a label map $f : V \rightarrow I$, the pullback f^*T is a k -uniform hypergraph on V whose edges are those k -tuples mapping into T . The incidence matrix gives a Hermitian bipartite weighted graph on $V \cup E(f^*T)$, and equitable partitions of this incidence graph recover the standard fractional/combinatorial CSP invariants. Lean: `Graphplay/Relational.lean`. For $k = 2$ this is the pullback graph of [amorc2025]; the $k > 2$ case appears not to have been treated through the equitable lens.

8. Toolkit Vision: Search/Transfer Compiler with Certificates

The previous note [amorc2025] described a Python prototype (`tools/search_compiler.py`) that takes a template and emits adjacency and Laplacian spectra, a CNO-style spectral-ratio diagnostic, and a quotient-Hamiltonian scan. With the spine in place we promote the prototype to a typed compiler whose signature is

$$\text{primitive} \times \text{hardware} \mapsto (\text{host}, \text{partition}, \text{schedule}, \text{certificate}).$$

The **primitive** is one of PST, uniform mixing, or spatial search; the **hardware** is a set of quotient constraints (which cells must exist; which couplings are realizable; which signings are realizable). The **host** is a `GraphBundle` (Section 3); the **partition** is an equitable partition of the host whose quotient encodes the primitive; the **schedule** is the (possibly time-dependent) Hamiltonian; the **certificate** is a term of a Lean type `Certificate primitive host schedule` whose inhabitation is exactly the relevant spine hypothesis (regularity, biregularity, signing phase-equitability, or graphon equitability).

Concretely, `Graphplay/Toolkit/Certificate.lean` will define structure `Certificate (p : Primitive) (B : Bundle) (S : Schedule) : Prop` with field constructors corresponding to Theorems 3.3, 7.1, and 5.4 as appropriate. The compiler emits both a JSON artifact and a Lean term; the Lean term is checked by `lake build` before deployment, giving a machine-verifiable bridge from a CSP-shaped engineering description to a quantum-walk schedule.

The case studies of [amorc2025] (`rook_3x3_equal_fiber`, `cycle8_powerlaw_alpha1`, `torus_heawood7_equal_fiber`) are precisely equal-fiber bundles whose templates carry verifiable regularity certificates; they become end-to-end Lean-checked instances of the compiler.

9. Connections and Open Directions

Graphops. Backhausz–Szegedy graphops generalize graphons by removing the L^2 symmetric-kernel assumption. The quotient spine survives when the operator commutes with the projection Π ; the graphon Theorem 5.5 should extend to graphops under mild compactness. Lean port open.

Surface topology. The Heawood map-color completion of [amorc2025] is now visible as the special bundle with $Q = K_{H(g)}$. The surface algebraic-connectivity ceiling of Freitas [freitas2001heawood] becomes a Tower-2 spectral budget on Q . A topology-aware bundle (where fibers carry a metric and Q is a triangulation) would unify surface coloring with the filtered-colimit picture.

Bose–Mesner versus filtered colimits. Conjecture: a family of finite weighted graphs admits a graphon quasi-infinite limit (Theorem 5.5) and a chiral signing speedup (Theorem 7.1) on a common partition if and only if the Bose–Mesner algebra of that partition coincides with the algebra generated by the characteristic projections of π . This would tie the chiral and graphon stories to the association-scheme literature in a clean way; it is consistent with [chan2019fractional] but open in general.

Open-system extensions. The bundle-and-quotient picture survives unitary perturbations whose generator is itself equitable. Lindbladian extensions, time-dependent schedules, and noisy CTQW are not yet treated.

References

Quantum-walk and spatial-search papers (Tamon corpus and immediate neighbors):

- *bachman2011quotient*. R. Bachman, E. Fredette, J. Fuller, M. Landry, M. Opperman, C. Tamon, A. Tollefson. Perfect state transfer on quotient graphs, arXiv:1108.0339.
- *ge2010products*. Y. Ge, B. Greenberg, O. Perez, C. Tamon. Perfect state transfer, graph products and equitable partitions, arXiv:1009.1340.

- *chan2022shadows*. A. Chan, C. Godsil, C. Tamon, W. Xie. Of shadows and gaps in spatial search, arXiv:2204.04355.
- *xie2023tail*. W. Xie, C. Tamon. No infinite tail beats optimal spatial search, arXiv:2301.07251.
- *bernard2022tails*. P.-A. Bernard, C. Tamon, L. Vinet, W. Xie. Quantum state transfer in graphs with tails, arXiv:2211.14704.
- *ide2022equitable*. Y. Ide, A. Narimatsu. Perfect state transfer, equitable partition and continuous-time quantum walk based search, arXiv:2209.07688.
- *chan2019fractional*. A. Chan, G. Coutinho, C. Tamon, L. Vinet, H. Zhan. Fractional revival and association schemes, arXiv:1907.04729.
- *kirkland2023join*. S. Kirkland, H. Monterde. Quantum walks on join graphs, arXiv:2312.06906.
- *levine2026chiral*. L. Levine, J. J. Mesapam, B. Mustico, C. Tamon, G. Tucker, H. Zhan. Uniform mixing in chiral quantum walks, arXiv:2605.04414.

Graphon foundations:

- *bick2024graphlimits*. C. Bick, D. Sclosa. Dynamical systems on graph limits and their symmetries, arXiv:2110.13686.
- *szegedy2010spectral*. B. Szegedy. Limits of kernel operators and the spectral regularity lemma, arXiv:1003.5588.
- *gao2020graphon*. S. Gao, P. E. Caines. Subspace decomposition for graphon LQR: applications to VLSNs of harmonic oscillators, arXiv:2004.00677.

Spatial-search references kept from v1:

- *chakraborty2020spatial*. S. Chakraborty, L. Novo, A. Ambainis, Y. Omar. Spatial search by quantum walk is optimal for almost all graphs, arXiv:1508.01327.
- *king2025longrange*. R. King et al. Optimal spatial searches with long-range tunneling, arXiv:2501.08148.
- *li2025deterministic*. X. Li, S. Luo, Z. Feng, Y. Li. Deterministic quantum search on all Laplacian integral graphs, arXiv:2506.21108.
- *sadowski2014planar*. P. Sadowski. Quantum spatial search on planar networks, arXiv:1406.0339.
- *wong2017lazy*. T. Wong. Faster search by lackadaisical quantum walk, arXiv:1706.06939.
- *freitas2001heawood*. M. A. A. de Freitas. A Heawood-type result for the algebraic connectivity of graphs on surfaces, arXiv:math/0109191.
- *amorc2025*. AMORC. *A.M.O.R.C. Toolkit Note: Engineered Color Templates for Spatial Search*. (Graphplay v1; this repository, paper/quasi_infinite_adjoint.typ.)